Kodavatikanti Bhuvan Chandra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.63
Narayana Junior College, Tirupati	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	98.50%
Narayana E.M HIGH SCHOOL, M.R.PALLI ,CHITTOOR DISTRICT	Class X (Board of Secondary Education, Andhra Pradesh)	2021	99.17%

EXPERIENCE

Oracle | Project Intern | Data Services Development

- Migrated the DICOM shared platform from **Spring Boot 2.x to 3.0.13 and Java 17**, aligning the DI ecosystem with **Spring Framework 6 and Jakarta EE** for modern framework compatibility.
- Aligned open-source and Oracle/internal dependencies by adopting the **Spring Boot BOM**, upgrading **Jersey 3 and Jetty 11**, resolving OCI SDK compatibility issues, and refreshing Gradle lockfiles for reproducible builds.
- Modernized application source code through **javax to jakarta** package migration, Spring Boot 3 API and configuration updates, dependency-injection changes, and Jersey/Jetty web-stack compatibility fixes.
- Validated migrated artifacts across downstream repositories through dependency updates, compilation, Gradle builds, and compatibility checks, establishing a **repeatable migration workflow** for future Spring Boot upgrades.

PROJECTS

Heatmap-Guided Probabilistic Human Pose Transition *(Research Project / Prof. Vineeth Balasubramanian)*

- Developed a probabilistic human pose transition framework modeling keypoint uncertainty through **top-K probable poses** and associated probabilities.
- Integrated heatmap-guided uncertainty, **PCA-based pose manifold constraints**, and variance-aware structural regularization to generate smooth, anatomically plausible transitions while suppressing noisy joints.
- Evaluated pose transitions using **PCPT and PCPK** metrics across **Yoga-31** source-target pairs with ViTPose heatmaps and BlazePose keypoints.

Document Management Repository (DMR) *(Coursework / Group Project / Prof. Manish Singh)* GitHub

- Developed a full-stack, AI-powered document management system using React, Node.js, Express, MongoDB, Redis, and **AWS S3**, supporting public, private, and organization-based document spaces.
- Integrated **LLaMA 3.3 70B** via Groq API for automated document tagging and AI-based vault routing, extracting domains, sensitivity levels, keywords, and confidence scores from uploaded documents.
- Secured document workflows with **JWT authentication**, role-based access control, granular permissions, tokenized sharing, zlib compression, Redis caching, and pre-signed S3 URLs.

StockML: Full-Stack Stock Price Prediction Platform GitHub

- Engineered stock price forecasting using **FastAPI, Next.js, Scikit-Learn, Pandas, and NumPy**, with OHLCV-based rolling features, an **MLP Regressor**, and chronological train-validation splitting.
- Added dynamic prediction caching and **RMSE-based anomaly detection**, integrating **Tavily** for contextual market news and an interactive dashboard for actual vs. predicted price visualization and anomaly analysis.

Event Ticket Booking System *(Coursework / Group Project / Prof. Shirshendu Das)* GitHub

- Developed a full-stack web application for movie and event ticket reservations with **real-time seat availability** and tiered pricing, implementing booking and cancellation logic with basic concurrency handling.
- Designed a **normalized SQL schema** and optimized seat availability lookups using indexing, with a user-friendly UI for online reservations and live seat status updates.

RISC-V Simulator *(Coursework / Group Project / Prof. Rajesh Kedia)* GitHub

- Built an assembler and simulator for the **RISC-V instruction set**, including cache simulation and cache usage statistics.
- Implemented a configurable **cache simulator** with adjustable parameters, tracking **hit/miss statistics** to analyze cache usage.

TEACHING

Probability & Random Variables - Undergraduate Teaching Assistant under Dr. Anjana Mahesh

Work Done: Evaluated answer scripts for over 100 students and led tutorial sessions with problem-solving support.

RELEVANT COURSES

AI: Foundations of Machine Learning, Deep Learning, Online Learning, Computer Vision, Natural Language Processing

CS: Computer Architecture, Discrete Mathematics, DBMS, Data Structures and Applications, Algorithms, Software Engineering

Mathematics: Probability and Random Variables, Calculus, Vector Calculus, Differential Equations, Applied Statistics

Electrical Engineering: Linear Systems and Signal Processing, Matrix Theory

SKILLS

- **Languages:** Python, C, C++, SQL, RISC-V Assembly
- **Libraries/ML:** NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTorch
- **Web Development:** HTML, CSS, JavaScript, FastAPI, React, Next.js, Django, Express.js
- **Tools:** Git, LaTeX, NetworkX, Linux
- **Competitive Programming:** LeetCode, Codeforces

POSITIONS OF RESPONSIBILITY

Clubs: Sunshine (Counselling Cell of IITH) -- UG Sunshine Buddy

Clubs: Sunshine (Counselling Cell of IITH) -- Newsletter Team

EXTRACURRICULAR

Volunteering: National Service Scheme (NSS), IITH -- Student Volunteer (186 hours)

Hobbies: Swimming, Table Tennis, Cricket, Reading Novels.